

**LAPORAN PRAKTIKUM
WORKSHOP KECERDASAN BUATAN
BAB 2**



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**PROGRAM STUDI SARJANA TERAPAN
TEKNOLOGI REKAYASA MULTIMEDIA
JURUSAN TEKNOLOGI MULTIMEDIA KREATIF
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
2026**

Example 2.1

```
#Example 2.1
print("Hello World!")
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_01.py"
Hello World!
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Example 2.2

```
#Example 2.2
'''
This is a multiple line
comments.
'''
print("Hello World")
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_02.py"
Hello World
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Example 2.3

```
#Example 2.3
a = 5
b = 7
c = a + b
print("c = " , c)
d = 'x'
e = 'y'
print(d + e)
f = "Hello"
g = "World"
l = f + " " + g
print(l)
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_03.py"
c = 12
xy
Hello World
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Example 2.4

```
#Example 2.4
for i in range(5):
```

```
print(i)
print("Finished")
```

```
GAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/
AI_02_04.py"
0
1
2
3
4
Finished
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.5

```
#Example 2.5
for i in range(5):
    for j in range(5):
        x = i * j
print(x)
print("Inner Loop Finished")
print("Outer Loop Finished")
```

```
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TU
GAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/
AI_02_05.py"
16
Inner Loop Finished
Outer Loop Finished
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.6

```
#Example 2.6
x = 0
while x < 5:
    x = x + 1
print(x)
print("Finished")
```

```
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TU
GAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/
AI_02_06.py"
5
Finished
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.7

```
#Example 2.7
x = 100
if x >= 70:
    print("Excellent")
elif x >= 60:
```

```
    print("Good")
else:
    print("OK")
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_07.py"
Excellent
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Example 2.8

```
#Example 2.8
def add (x,y):
    z = x + y
    return z
t = add(30, 20)
print(t)
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_08.py"
50
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Example 2.9

```
#Example 2.9
def maxarray (xs):
    m = xs[0]
    for x in xs:
        if m < x:
            m = x
    return m
data = [0,1,2,3,4,5]
t = maxarray(data)
print(t)
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_09.py"
5
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Exercise 2.9.1

```
# Exercise 2.9.1

def minarray(xs):
    m = xs[0]
    for x in xs:
        if m > x:
```

```

        m = x
    return m

data = [0, 1, 2, 3, 4, 5]
t = minarray(data)
print(t)

```

```

PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/AI BAB 2/CODE/AI_02_09.01.py"
0
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>

```

Exercise 2.9.2

```

# Exercise 2.9.2

def sortarray(xs):
    xs.sort()
    return xs

data = [5, 2, 4, 1, 3, 0]
t = sortarray(data)
print(t)

```

```

PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/AI BAB 2/CODE/AI_02_09.02.py"
[0, 1, 2, 3, 4, 5]
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>

```

Example 2.10

```

#Example 2.10
print('What is your name? ')
x = input()
print('Hello ' + x + "!")

```

```

(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_10.py"
What is your name?
pandu
Hello pandu!
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>

```

Example 2.11

```

#Example 2.11
print('Input a number: ')
x = input()
y = int(x)
y = y ** 2

```

```
print('The square of ' + str(x) + ' is ' + str(y))
```

```
● (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_11.py"
Input a number:
23
The square of 23 is 529
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Exercise 2.11.1

```
# Exercise 2.11.1

print('Input a float number: ')
x = input()
y = float(x)
y = y ** 2
print('The square of ' + str(x) + ' is ' + str(y))
```

```
● PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/AI BAB 2/CODE/AI_02_11.01.py"
Input a float number:
23
The square of 23 is 529.0
○ PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> █
```

Exercise 2.11.2

```
# Exercise 2.11.2

print('Input your name: ')
name = input()

print('Input your age: ')
age = int(input())

print('Input your marks: ')
marks = float(input())

print('Name: ' + name)
print('Age: ' + str(age))
print('Marks: ' + str(marks))
```



```
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/AI BAB 2/CODE/AI_02_11.02.py"
pandu
Input your age:
20
Input your marks:
100
Name: pandu
Age: 20
Marks: 100.0
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.12

```
#Example 2.12
s = input("Input a list of numbers: ")
numbers = list(map(int, s.split()))
print(numbers)
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_12.py"
Input a list of numbers: 23
[23]
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.13

```
#Example 2.13
import math
r = 5
cir = 2 * math.pi * r
print('The circumference is ' + str(cir))
x = 5
y = math.sqrt(x)
print("The square root of " + str(x) + " is " + str(y))
x = 7
y = math.factorial(x)
print("The factorial of " + str(x) + " is " + str(y))
x = 16.4
y = math.floor(x)
print("The floor of " + str(x) + " is " + str(y))
x = 16.4
y = math.ceil(x)
print("The ceiling of " + str(x) + " is " + str(y))
```

```
• (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_13.py"
The circumference is 31.41592653589793
The square root of 5 is 2.23606797749979
The factorial of 7 is 5040
The floor of 16.4 is 16
The ceiling of 16.4 is 17
○ (venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Exercise 2.13.1

```
# Exercise 2.13.1

import math

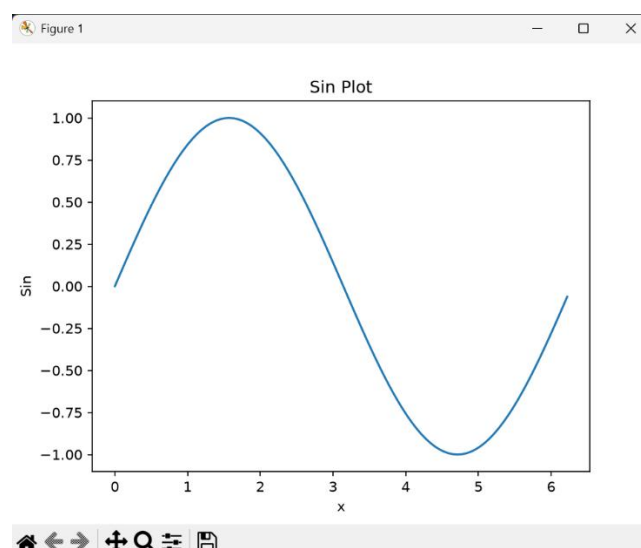
print('Input a list of float numbers:')
numbers = input().split()

for x in numbers:
    x = float(x)
    y = math.sin(x)
    print('The sine of ' + str(x) + ' is ' + str(y))
```

```
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/AI BAB 2/CODE/AI_02_13.01.py"
Input a list of float numbers:
1.2
The sine of 1.2 is 0.9320390859672263
PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

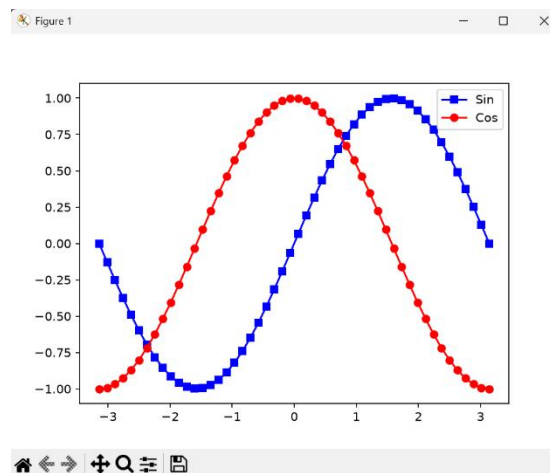
Example 2.14

```
#Example 2.14
import matplotlib.pyplot as plt
import numpy as np
x = [i for i in range(100)]
x = np.array(x)
x = 2 * np.pi * x * 0.01
y = np.sin(x)
plt.plot(x,y)
plt.title('Sin Plot')
plt.xlabel('x')
plt.ylabel('Sin')
plt.show()
```



Example 2.15

```
#Example 2.15
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-np.pi, np.pi, 50)
y1 = np.sin(x)
y2 = np.cos(x)
plt.plot(x, y1, color = 'blue', marker = "s", label='Sin')
plt.plot(x, y2, color = 'red', marker = "o", label='Cos')
plt.legend()
plt.show()
```



Exercise 2.15.1

```
# Exercise 2.15.1

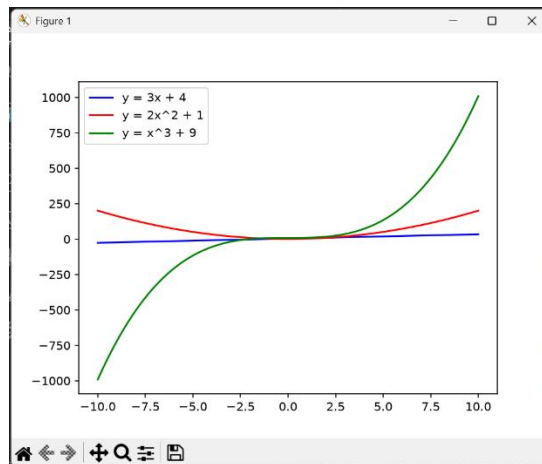
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-10, 10, 100)

y1 = 3 * x + 4
y2 = 2 * x**2 + 1
y3 = x**3 + 9

plt.plot(x, y1, color='blue', label='y = 3x + 4')
plt.plot(x, y2, color='red', label='y = 2x^2 + 1')
plt.plot(x, y3, color='green', label='y = x^3 + 9')

plt.legend()
plt.show()
```



Example 2.16

```
#Example 2.16
import pandas as pd
data = {'Name': ['Tony', 'Robert', 'John', 'Alice'],
        'Age': [18, 24, 19, 21],
        }
df = pd.DataFrame (data, columns = ['Name', 'Age'])
print (df)
df.to_csv('test.csv', index=False)
```

```
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:/Users/Administrator/Documents/TUGAS/Semester 5/AI/CODE/AI_02_16.py"
      Name  Age
0    Tony   18
1  Robert   24
2   John   19
3  Alice   21
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> 
```

Example 2.17

```
#Example 2.17
import pandas
df = pandas.read_csv('test.csv')
print (df)
```

```
CODE > test.csv
1 Name, Age
2 Tony, 18
3 Robert, 24
4 John, 19
5 Alice, 21
6

Problems Output Debug Console Terminal Ports
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI> & "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\venv_cv\Scripts\python.exe" "c:\Users\Administrator\Documents\TUGAS\Semester 5\AI\CODE\AI_02_17.py"
      Name  Age
0      Tony   18
1    Robert   24
2      John   19
3      Alice   21
(venv_cv) PS C:\Users\Administrator\Documents\TUGAS\Semester 5\AI>
```

Example 2.18

```
#Example 2.18
#pip install xlswriter
import pandas as pd
data = {'Name': ['Tony', 'Robert', 'John', 'Alice'],
        'Age': [18, 24, 19, 21],
        }
df = pd.DataFrame (data, columns = ['Name', 'Age'])
print (df)
writer = pd.ExcelWriter("test.xlsx", engine='xlswriter')
#df.to_excel(writer, sheet_name='Sheet1',
# startrow=7, startcol=4, header=False, index=False)
df.to_excel(writer, sheet_name = 'Sheet1', index=False)
writer.close()
```

	A	B	C	D	E	F	G
1	Name	Age					
2	Tony	18					
3	Robert	24					
4	John	19					
5	Alice	21					
6							
7							
8							
9							
10							
11							

Example 2.19

```
#Example 2.19
#pip install xlswriter
import pandas
df = pandas.read_excel('test.xlsx', sheet_name='Sheet1')
print(df)
```

